

AI-ENHANCED MOTION CONTROL FOR DIGITAL HYDRAULIC LINEAR ACTUATORS

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ABSTRACT

This study evaluates a model-guided, human-in-the-loop method for revising a fuzzy logic controller for a physical Digital Hydraulic Linear Actuator (DHLA) driven by pulse-width-modulated on/off valves. The original operator-generated controller, Profile 1, was compared with two large-language-model-assisted revisions, Profiles 2 and 3. The language model reviewed the original LabVIEW fuzzy profile, an empirical actuator model, one baseline simulation, five physical runs per profile, and extracted performance metrics, then proposed membership function and rule-mapping changes that were manually implemented in LabVIEW. Under the evaluated 125-s reference trajectory, Profile 2 reduced cumulative absolute error from 515.814 to 411.781 cm·s, reduced mean run-level worst overshoot from 55.994% to 43.225%, and reduced final absolute error from 2.273 to 1.965 cm relative to Profile 1, although extension and retraction velocities were lower. Profile 3 increased extension velocity 21.34% relative to Profile 2 as well as increased overshoot, final error, and valve usage. The empirical simulation reproduced final-error ordering but it inverted the velocity and overshoot rankings, and did not predict the physical tradeoff. The results support using a language model as an engineering design assistant while confirming that manual implementation, objective weighting, and physical validation remain necessary.

Keywords: AI-assisted fuzzy control, Digital Hydraulic Linear Actuator (DHLA), human-in-the-loop controller tuning, PWM valve control, empirical hydraulic modeling

1. INTRODUCTION

Hydraulic actuators provide high power density and environmental robustness, but precise motion control is complicated by fluid compressibility, temperature-dependent viscosity, valve dynamics, friction, and load variation. Digital hydraulic systems use binary on/off valves and pulse-width modulation (PWM) to approximate proportional flow. This architecture can reduce throttling losses, but switching delay, pressure

transients, and quantized valve authority complicate trajectory tracking. Passivity-based control and integrated smart-manifold approaches have been developed for high-bandwidth hydraulic actuation [1, 2], while robust sliding-mode methods have been investigated for integrated hydromechatronic actuators [3].

Fixed-gain linear controllers can lose tracking accuracy when hydraulic dynamics vary outside the conditions used for tuning. Fuzzy Logic Control (FLC) is one practical alternative because it can encode nonlinear, asymmetric control actions through interpretable membership functions and rules [4–7].

The performance of an FLC depends strongly on the placement and shape of its membership functions and on the rule mapping between position error and valve command. Manual tuning can be time-consuming and may produce controller settings that are specific to one test condition. Particle Swarm Optimization has been used to automate fuzzy-controller tuning for electro-hydraulic actuators [8], and recent work has examined reinforcement-learning-based hydraulic tracking controllers [9–11]. These methods demonstrate the broader value of data-informed controller tuning, but they differ substantially from the heuristic large-language-model-assisted workflow evaluated in this paper.

This work evaluates whether a large language model can serve as a model-guided engineering design assistant for revising a distance-only fuzzy controller while preserving the existing DHLA hardware and control architecture. Two AI-assisted controller iterations are manually implemented and compared with the original controller through repeated physical testing and empirical simulation. The contribution is a documented human-in-the-loop tuning workflow and a physical comparison showing both an improvement and a subsequent speed-versus-containment tradeoff.

2. SYSTEM DESIGN

The system consists of a hydraulic cylinder, two high-speed normally open poppet valves contained in a custom designed

manifold, and a rectangular pump/reservoir unit. PWM signals are generated by a personal computer running LabVIEW and transmitted through a National Instruments data acquisition board. The physical test setup and forward-stroke operating case are shown in Figs. 1 and 2.

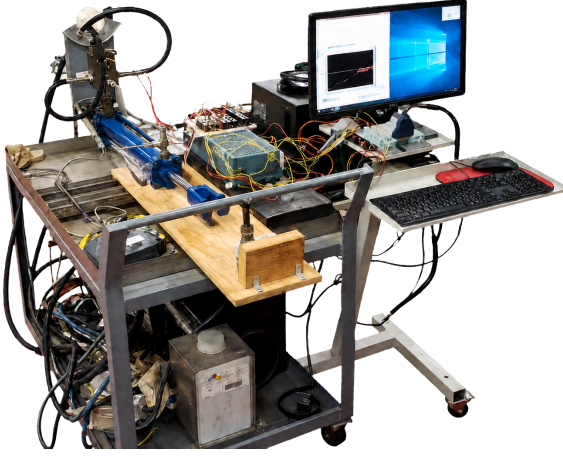


FIGURE 1: ROBOTIC HYDRAULIC TEST INSTRUMENTATION SETUP.

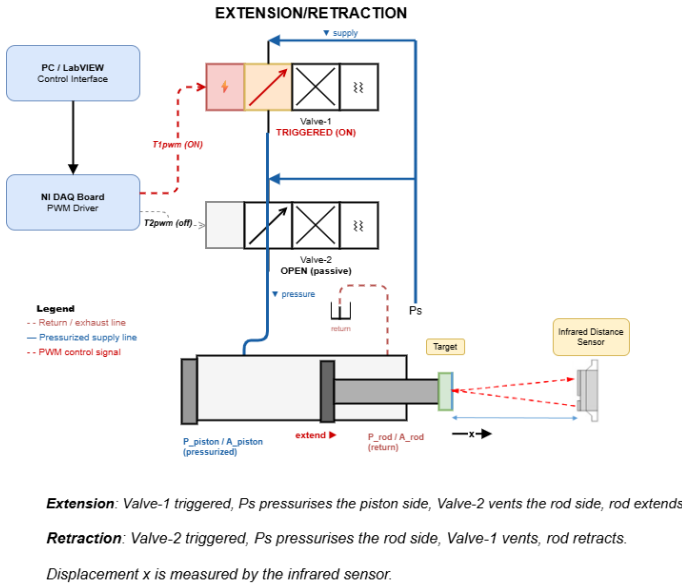


FIGURE 2: FORWARD STROKE OPERATION CASE.

The solenoid valves open or close the hydraulic lines and are rated for pressures up to 4000 psi. Measured opening and closing times are summarized with the cylinder and fluid specifications in Table 1. The pump is capable of delivering approximately 50 psi; the cylinder and valve pressure values in Table 1 are component ratings. The actual operating pressure used during the reported tests was not recorded.

TABLE 1: SYSTEM PARAMETERS

Component	Specification
Cylinder	Double acting; 2-in bore; 14-in stroke; 1.125-in rod diameter; 2500-psi maximum rated pressure
Fluid	API gravity at 60 °F: 22.0; specific gravity at 60 °F: 0.9218; viscosity at 40 °C: 32.0 cSt
Trigger valve	Solenoid on/off; normally open; $T_{on} = 1.84$ ms; $T_{off} = 1.92$ ms; 4000-psi maximum rated pressure

A custom-designed manifold block was used to reduce packaging volume, weight, and leakage points relative to individually housed valves. The LabVIEW controller generated the pulsed signals $T1_{pwm}$ and $T2_{pwm}$ for the upper and lower valves. Two 3D-printed components were added to improve position measurement repeatability. A flat sensor target face attached to the piston rod reduced variation caused by the original curved reflective surface, and an anti-rotation foot maintained the target face alignment with the infrared distance sensor throughout the stroke. These modifications affected measurement quality but did not change the hydraulic circuit or control electronics. The actuator, valve hardware, LabVIEW interface, and measurement arrangement were unchanged across Profiles 1, 2, and 3, allowing direct comparison between the operator-generated fuzzy baseline and the two AI-assisted fuzzy revisions.

3. EXPERIMENTAL SETUP

The study used a sequential, human-in-the-loop workflow: (1) collect baseline physical data using Profile 1; (2) identify a reduced empirical actuator model; (3) provide the controller files, model, run data, and metrics to the language model; (4) manually implement the proposed Profile 2 revision in LabVIEW and test it; (5) provide the Profile 2 results for a second Profile 3 revision; and (6) compare all profiles using the same analysis workflow. The language model did not execute the controller, modify LabVIEW directly, or perform autonomous online adaptation.

3.1. Baseline Controller and Recorded Signals

Profile 1 was the manually generated fuzzy controller. Its only fuzzy input was position error,

$$e_k = r_k - x_k, \quad (1)$$

where r_k is desired position and x_k is measured position at sample k . The controller produced upper- and lower-valve PWM duty-cycle commands. During each run, time, desired position, measured position, upper-valve command, and lower-valve command were recorded.

Each physical run lasted approximately 125 s and contained about 4061 samples. The median recorded interval was 0.029982 s, corresponding to approximately 33.35 Hz. The valves were driven using 50-Hz PWM; the physical controller-loop frequency was not separately recorded. Five runs were completed for each profile using the same predefined desired-position trajectory shown in Figs. 4, 6, and 8. The trajectory spanned 4–32 cm and included 4cm staircase changes and larger reversals, with most targets held for approximately 4 s. Testing proceeded in the order Profile 1, Profile 2, and Profile 3. Runs within each profile group were separated by approximately 1–2 min, and the machine was allowed to cool fully before the next profile group. No external load was applied. Operating hydraulic pressure and oil temperature were not recorded.

3.2. Empirical Dynamic Model and Simulation

The LabVIEW simulation used the reduced empirical acceleration model

$$a_k = K_{upper} u_{upper,k} - K_{lower} u_{lower,k} - cv_k, \quad (2)$$

where $u_{upper,k}$ and $u_{lower,k}$ are valve duty-cycle commands, $upper$ and $lower$ are experimentally estimated acceleration gains, c is a damping coefficient, and v_k is piston velocity. Acceleration was numerically integrated to velocity and position. The model replaces an explicit pressure-force calculation with calibrated gains and therefore represents a reduced second order approximation rather than a complete hydraulic model.

The coefficients were estimated from five physical calibration runs using Python. Position was smoothed with a 21-sample moving average, velocity and acceleration were calculated by finite differences, and normalized valve commands were fit by linear least squares. Commands were normalized between the recorded closed command of 0.01 and full command of 0.973014. The five-run estimates were $upper = 13.691 \pm 0.204 \text{ cm/s}^2$, $K_{lower} = 14.817 \pm 0.150 \text{ cm/s}^2$, and $c = 1.936 \pm 0.083 \text{ s}^{-1}$, with mean $R^2 = 0.654 \pm 0.009$. Representative values based on these averages were selected for the LabVIEW simulation as $K_{upper} = 15 \text{ cm/s}^2$, $K_{lower} = 15 \text{ cm/s}^2$, and $c = 1.8 \text{ s}^{-1}$, giving $a_k = 15u_{upper,k} - 15u_{lower,k} - 1.8v_k$. The simulation used a 0.010-s integration step.

The simulation was used to screen controller revisions before physical implementation. It did not explicitly represent chamber pressure, pressure-flow nonlinearities, valve dead time, stiction, leakage, hose compliance, pressure transients, or sensor quantization. These omissions are relevant when interpreting differences between simulated and physical behavior.

3.3. AI-Assisted Fuzzy-Controller Revision

The controller-design package outlined in Appendix A, was supplied to ChatGPT-5.5 Thinking through a ChatGPT Plus account in April–May 2026. This package contained: the original LabVIEW fuzzy profile in PDF and .fs formats, one baseline simulation run, five Profile 1 physical runs, a consolidated performance-metric file, a file describing the supplied materials, the empirical model in Eq. 2, and two tuning prompts. Prompt 1 established a model-guided diagnostic role focused on second order dynamics, asymmetry, damping, overshoot, and controller limitations. Prompt 2 requested a complete revised fuzzy profile and prioritized tracking error, overshoot, settling, velocity, and valve authority. The complete prompts used for the two controller revision iterations are provided in Appendices B and C.

The tuning process was a sequential heuristic search, not regression, gradient descent, Particle Swarm Optimization, a Genetic Algorithm, ANFIS training, reinforcement learning, or an exhaustive nonlinear optimizer. ChatGPT proposed a complete membership-function and rule-map revision, the team manually implemented it in LabVIEW, physical and simulated results were calculated, and those results informed the next revision. Table 2 summarizes the three controller iterations.

TABLE 2: AI-ASSISTED FUZZY-CONTROLLER ITERATIONS

Iteration	Profile	Main change	Physical result
0	Profile 1	Original five-input-region, three-output-region controller	Baseline
1	Profile2	Seven input regions and four output regions; full-open output shifted near 0.99	Best physical balance
2	Profile3	Wider accurate region and increased medium valve outputs	Faster than Profile 2, but more overshoot

The design loop used the following procedure:

1. Evaluate the current profile using cumulative error, overshoot, velocity, final error, settling behavior, and valve usage.
2. Diagnose weak error regions, insufficient valve authority, or excessive near-target correction using the empirical model and physical results.
3. Propose revised input and output membership functions and the associated rule mapping.
4. Manually implement the revised profile in the LabVIEW fuzzy editor.
5. Run the profile in simulation and on the physical actuator.
6. Compare the new results with the prior profile and retain or reject the revision using physical performance as the final criterion.

3.4. Physical Implementation and Test Consistency

The original one-input, two-output control architecture was preserved. No velocity input, pressure feedback, or additional sensor signal was added. Profiles 2 and 3 were entered manually into the LabVIEW fuzzy editor and tested on the same actuator, sensor arrangement, valve manifold, NI DAQ interface, and reference trajectory used for Profile 1. This isolates the comparison primarily to the membership functions and rule mapping.

3.5. Performance Metrics and Data Processing

The cumulative absolute error was calculated as

$$J_e = \sum_{k=1}^N |e_k| \Delta t, \quad (3)$$

and final absolute error was $|e_N|$. Valve usage was represented by

$$J_u = \sum_{k=1}^N (u_{upper,k} + u_{lower,k}) \Delta t. \quad (4)$$

For a transition from r_0 to r_1 , percentage overshoot was calculated relative to commanded step magnitude as

$$M_p = \frac{|x_{peak} - r_1|}{|r_1 - r_0|} \times 100\%, \quad (5)$$

with x_{peak} selected in the direction beyond the new target. Overshoot was evaluated for each 4-cm target transition. The worst qualifying transition was identified separately for each physical run, and the five run-level values were averaged. For each simulation profile, the reported value is the worst 4-cm transition in the single simulated run. Velocity was calculated from position by finite differences and smoothed before the maximum positive and negative values were extracted. These smoothed peaks are reported as robust maximum extension and retraction velocities to reduce sensitivity to isolated distance-sensor quantization or jumps.

The physical performance values are descriptive aggregates from five runs per profile. Because this was a pilot comparison with $n = 5$ per controller, no inferential hypothesis tests were applied.

4. EXPERIMENTAL RESULTS AND DISCUSSION

This section compares the operator-generated Profile 1 controller with the AI-assisted Profile 2 and Profile 3 revisions. Five physical runs were analyzed for each profile. The main comparison metrics were robust maximum extension and retraction velocity, cumulative absolute error, mean run-level worst overshoot, final absolute error, maximum position, and valve usage.

4.1. Aggregate Physical Performance

TABLE 3: PHYSICAL AGGREGATE PERFORMANCE METRICS

Profile	Runs	Robust Max Ext. Vel. (cm/s)	Robust Max Ret. Vel. (cm/s)	Cum. Abs. Error (cm·s)	Mean Worst Overshoot (%)	Final Abs. Error (cm)
Profile 1	5	15.556	15.624	515.814	55.994	2.273
Profile 2	5	9.615	13.233	411.781	43.225	1.965
Profile 3	5	11.667	13.527	412.644	46.066	2.041

Table 3 shows that Profile 1 produced the highest velocities but also the highest cumulative error and worst overshoot. Profile 2 reduced cumulative error by 20.17%, worst overshoot by 22.80%, and final absolute error by 13.55% relative to Profile 1. Profile 3 recovered extension speed relative to Profile 2 but did not improve total tracking performance.

4.2. Profile 1 Baseline Behavior

Profile 1 used five position-error membership functions and three output membership functions for each valve. Table 4 directly pairs each input region with its valve-output rule, and Fig. 3 shows the corresponding membership functions.

TABLE 4: PROFILE 1 DISTANCE AND DUTY-CYCLE MEMBERSHIP FUNCTIONS WITH RULE MAPPING

I/O Variable	Membership function	Shape	Distance from desired and valve opening
I distance	Far Negative	Trapezoid	-27, -27, -12.5, -11
O duty-cycle-top	Open		0.6, 0.75, 0.99, 0.99
I distance	Close Negative	Gaussian	-14, -12, -11, -1
O duty-cycle-top	Slightly Open		0.05, 0.6, 0.6, 0.65
I distance	Accurate	Trapezoid	-0.75, -0.75, 0.75, 0.75
O duty-cycle-top	Closed		0.01, 0.01, 0.01, 0.025
O duty-cycle-bottom	Closed		0.01, 0.01, 0.01, 0.025
I distance	Close Positive	Gaussian	1, 10, 12, 14
O duty-cycle-bottom	Slightly Open		0.05, 0.6, 0.6, 0.65
I distance	Far Positive	Trapezoid	11, 12.5, 27, 27
O duty-cycle-bottom	Open		0.6, 0.75, 0.99, 0.99

Negative error commanded the top valve, positive error commanded the bottom valve, and the accurate region closed both valves. The five-region input structure provided limited separation between full drive and near-target correction.

Profile 1 produced robust extension and retraction velocities of 15.556 and 15.624 cm/s, but cumulative absolute error was 515.814 cm·s and worst overshoot was 55.994%. The controller used position error only, so it could not directly detect piston momentum. Once velocity accumulated, valve closure after entry into the near-target region could not immediately stop the piston. Valve switching delay, integration of acceleration into velocity and position, asymmetric actuator dynamics, and Center-of-Area output centroids all contributed to target crossing. Percentage overshoot can also become large when the peak displacement is normalized by a relatively small commanded step.

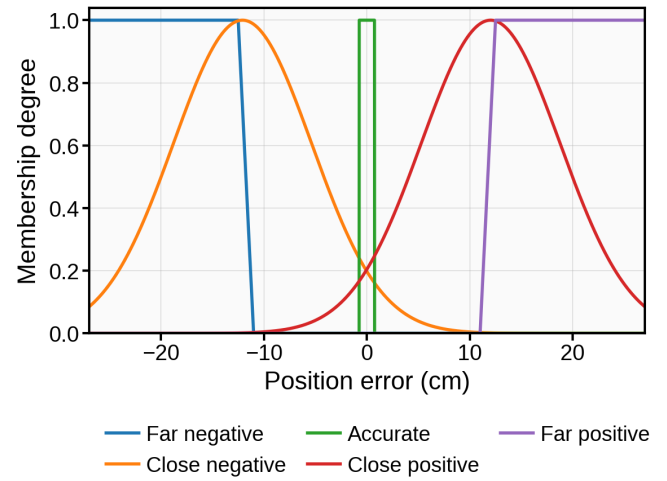


FIGURE 3: PROFILE 1 INPUT MEMBERSHIP FUNCTIONS.

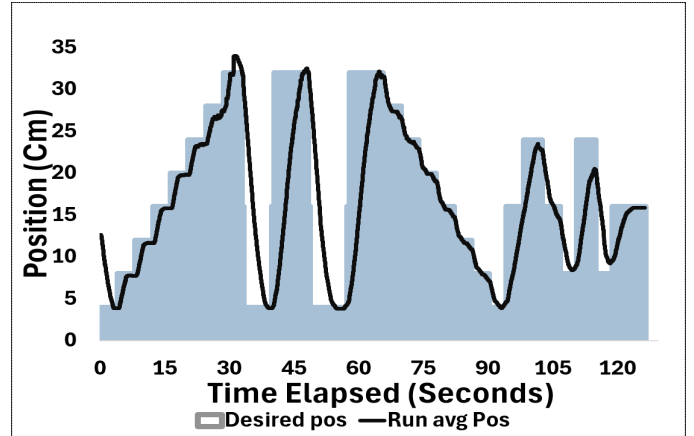


FIGURE 4: PROFILE 1 FIVE-RUN AVERAGE RESPONSE AND COMMANDED TRAJECTORY.

4.3. Profile 2 Physical Performance

Profile 2 expanded the distance input from five to seven regions and each valve output from three to four regions. Table 5 lists the resulting distance and duty-cycle sets and their rule mapping, while Fig. 5 shows the corresponding membership functions.

The full-open output region was shifted near the maximum duty-cycle range to address the low Center-of-Area centroid of the original output set, while medium and slight-output regions allowed staged reduction of valve authority near the target.

Profile 2 produced the best physical containment. Cumulative absolute error decreased to 411.781 cm·s, worst overshoot decreased to 43.225%, and final absolute error decreased to 1.965 cm.

Extension velocity decreased to 9.615 cm/s, but the lower speed reduced accumulated momentum before target crossing. The average maximum position was approximately 32.559 cm.

4.4. Profile 3 Physical Performance

Profile 3 retained the seven-region input structure, widened the accurate region, and increased medium-output duty-cycle authority to recover speed outside the near-target band. Table 6 lists the revised distance and duty-cycle definitions, and Fig. 7 shows the resulting membership functions.

TABLE 5: PROFILE 2 DISTANCE AND DUTY-CYCLE MEMBERSHIP FUNCTIONS WITH RULE MAPPING

I/O Variable	Membership function	Shape	Distance from desired and valve opening
I distance	Far Negative	Trapezoid	-27, -27, -9.5, -6.5
O duty-cycle-top	Open		0.94, 0.985, 0.99, 0.99
I distance	Med Negative	Trapezoid	-8, -5.5, -2, -1
O duty-cycle-top	Med Open		0.36, 0.52, 0.7, 0.84
I distance	Small Negative	Trapezoid	-1.8, -1, -0.45, -0.2
O duty-cycle-top	Slightly Open		0.06, 0.14, 0.24, 0.34
I distance	Accurate	Trapezoid	-0.35, -0.15, 0.15, 0.35
O duty-cycle-top	Closed		0.01, 0.01, 0.012, 0.03
O duty-cycle-bottom	Closed		0.01, 0.01, 0.012, 0.03
I distance	Small Positive	Trapezoid	0.2, 0.45, 1, 1.7
O duty-cycle-bottom	Slightly Open		0.05, 0.11, 0.18, 0.28
I distance	Med Positive	Trapezoid	0.9, 1.8, 4.8, 6.5
O duty-cycle-bottom	Med Open		0.28, 0.42, 0.56, 0.7
I distance	Far Positive	Trapezoid	4.8, 6.8, 27, 27
O duty-cycle-bottom	Open		0.94, 0.985, 0.99, 0.99

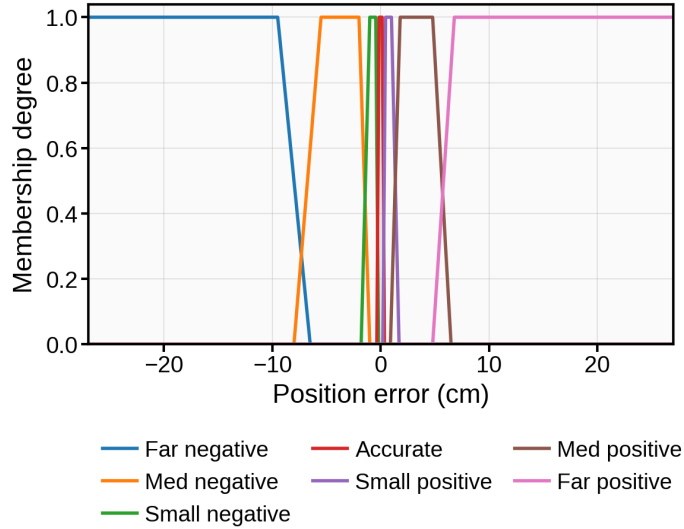


FIGURE 5: PROFILE 2 INPUT MEMBERSHIP FUNCTIONS.

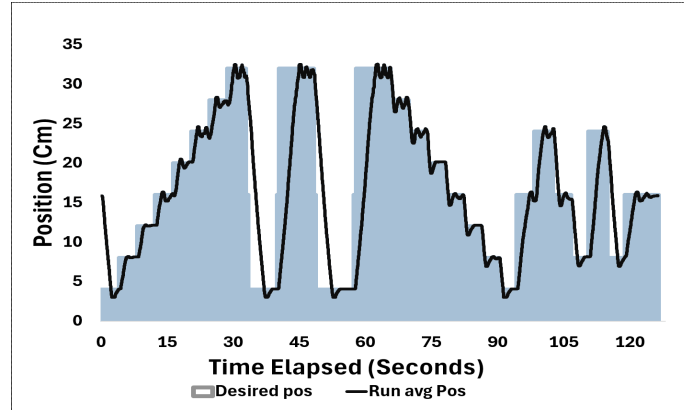


FIGURE 6: PROFILE 2 FIVE-RUN AVERAGE RESPONSE AND COMMANDED TRAJECTORY.

Profile 3 increased extension velocity from 9.615 to 11.667 cm/s relative to Profile 2, but worst overshoot increased from 43.225% to 46.066%, final absolute error increased from 1.965 to 2.041 cm, and cumulative error remained essentially unchanged at 412.644 cm·s. The average maximum position increased to approximately 34.052 cm. Profile 3 therefore recovered speed but did not improve overall physical tracking.

TABLE 6: PROFILE 3 DISTANCE AND DUTY-CYCLE MEMBERSHIP FUNCTIONS WITH RULE MAPPING

I/O Variable	Membership function	Shape	Distance from desired and valve opening
I distance	Far Negative	Trapezoid	-27, -27, -8.2, -6.2
O duty-cycle-top	Open		0.94, 0.985, 0.99, 0.99
I distance	Med Negative	Trapezoid	-6.6, -4.9, -2.1, -1.05
O duty-cycle-top	Med Open		0.64, 0.76, 0.88, 0.94
I distance	Small Negative	Trapezoid	-1.3, -0.9, -0.62, -0.42
O duty-cycle-top	Slightly Open		0.14, 0.26, 0.34, 0.42
I distance	Accurate	Trapezoid	-0.55, -0.32, 0.32, 0.55
O duty-cycle-top	Closed		0.01, 0.01, 0.015, 0.035
O duty-cycle-bottom	Closed		0.01, 0.01, 0.015, 0.035
I distance	Small Positive	Trapezoid	0.42, 0.62, 0.9, 1.3
O duty-cycle-bottom	Slightly Open		0.11, 0.23, 0.30, 0.38
I distance	Med Positive	Trapezoid	0.95, 1.75, 4.3, 5.6
O duty-cycle-bottom	Med Open		0.60, 0.72, 0.84, 0.92
I distance	Far Positive	Trapezoid	4.3, 6, 27, 27
O duty-cycle-bottom	Open		0.94, 0.985, 0.99, 0.99

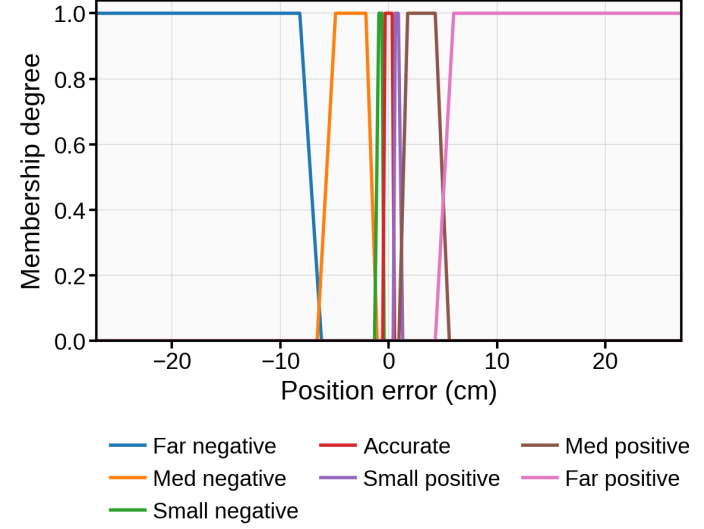


FIGURE 7: PROFILE 3 INPUT MEMBERSHIP FUNCTIONS.

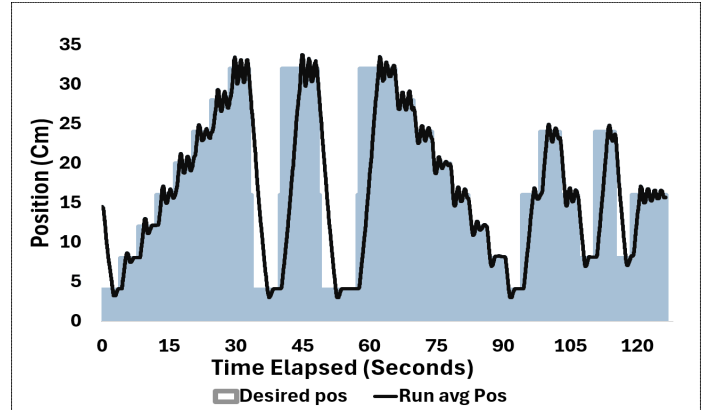


FIGURE 8: PROFILE 3 FIVE-RUN AVERAGE RESPONSE AND COMMANDED TRAJECTORY.

4.5. Physical Tradeoff and Directional Asymmetry

Table 7 quantifies the Profile 2-to-Profile 3 speed-versus-containment tradeoff. Retraction was faster than extension for Profiles 2 and 3. The physical coefficient estimates also showed a larger mean lower-valve gain than upper-valve gain, whereas the implemented simulation used equal representative gains of 15 cm/s². Different effective piston areas and other unmodeled hydraulic effects may contribute to the measured directional asymmetry. Future profiles should focus on asymmetric tune extension and retraction outputs.

TABLE 7: PROFILE 3 VERSUS PROFILE 2 PHYSICAL TRADE-OFF

Metric	Profile 2	Profile 3	Change
Extension velocity (cm/s)	9.615	11.667	+21.34%
Retraction velocity (cm/s)	13.233	13.527	+2.22%
Cumulative error (cm·s)	411.781	412.644	+0.21%
Worst overshoot (%)	43.225	46.066	+6.57%
Final absolute error (cm)	1.965	2.041	+3.87%
Maximum position (cm)	32.559	34.052	+4.58%
Valve usage integral	43.326	57.909	+33.66%

4.6. Simulation Versus Physical Results

TABLE 8: SIMULATION PERFORMANCE METRICS

Profile	Test Type	Robust Max Ext. Vel. (cm/s)	Robust Max Ret. Vel. (cm/s)	Cum. Abs. Error (cm·s)	Worst 4-cm Overshoot (%)	Final Abs. Error (cm)
Profile 1	Simulation	6.720	6.735	534.473	14.048	2.285
Profile 2	Simulation	7.965	7.985	406.668	20.853	1.849
Profile 3	Simulation	7.900	7.910	394.579	28.183	1.924

Simulation predicted lower cumulative error for Profile 3 than Profile 2, 394.579 versus 406.668 cm·s, but Profile 2 retained the lower simulated overshoot, 20.853% versus 28.183%. Thus, simulation did not identify a single profile as superior across all objectives.

Table 8 provides the corresponding simulation metrics. The physical tests showed a similar tradeoff direction but a different magnitude. Profile 3 provided higher physical extension velocity than Profile 2, yet its overshoot and valve usage increased. The reduced model was useful for screening possible controller changes, but missing valve delay, pressure-flow nonlinearities, stiction, leakage, hose compliance, pressure transients, and sensor quantization limited quantitative transfer to the physical actuator.

To improve concision and reduce redundant tables, overshoot values are reported directly in Tables 3 and 8.

4.7. Prompt Refinement and Study Limitations

The Profile 2-to-Profile 3 comparison demonstrates that prompt objectives act as a control-design variable. Table 9 summarizes the priorities recommended for future tuning. The second revision encouraged stronger valve authority and velocity without imposing a sufficiently strong penalty when speed increased overshoot or valve usage. Future prompts should prioritize cumulative error, overshoot, and settling behavior above velocity, and should permit increased valve authority only when physical tracking improves.

TABLE 9: RECOMMENDED WEIGHTED OBJECTIVES FOR FUTURE TUNING

Objective	Priority	Reason
Reduce cumulative absolute error	High	Measures total tracking burden
Reduce overshoot	High	Limits target crossing and mechanical risk
Improve settling behavior	High	Indicates stable convergence
Reduce final absolute error	Medium-high	Improves target holding
Increase velocity	Medium	Beneficial only when containment is maintained
Reduce valve usage	Medium	Limits unnecessary switching

The study is limited to five runs per profile, one actuator, one reference trajectory, and no-load testing. Operating pressure and oil temperature were not recorded, and profile order was not randomized. Full machine cooldown was used between profile groups, but equivalent thermal conditions could not be quantified. The controller used position error only, and the empirical model did not reproduce all hydraulic effects. These limitations restrict generalization but do not invalidate the observed within-platform comparison.

4.8. Overall Result

Profile 2 produced the best physical balance among the three tested profiles under the evaluated conditions. It minimized cumulative error, overshoot, and final error, while Profile 3 demonstrated that recovering speed through stronger valve commands can reduce containment. The results support physical validation as the final decision criterion for model-guided controller revisions.

5. CONCLUSION

This study evaluated a model-guided, human-in-the-loop method for using a large language model to revise the membership functions and rule mapping of a position-error-based fuzzy controller for a physical DHLA. Relative to the manually generated Profile 1 controller, Profile 2 reduced cumulative absolute error, mean run-level worst overshoot, and final absolute error, although at reduced extension and retraction velocity. A second AI-assisted revision, Profile 3, recovered extension speed but increased overshoot, final error, and valve usage, demonstrating the importance of weighting stability and tracking objectives above unconstrained speed. The simulation supported preliminary screening but did not fully predict physical behavior. These findings support the use of language models as engineering design assistants while emphasizing that manual implementation, transparent metric definitions, and physical validation remain necessary. Future work should report run-level uncertainty, validate the empirical model over broader operating conditions, incorporate velocity or pressure information, and evaluate formally defined multi-objective tuning methods.

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APPENDIX A

FILES PROVIDED TO THE CHATBOT

The following files were provided to the chatbot to evaluate Fuzzy Profile 1 and tune the fuzzy controller:

- [1] Explanation of Files.txt: Manifest explaining the controller files, recorded test data, baseline metrics, and use of Prompt1 and Prompt2.
- [2] piston fuzzy control.fs: LabVIEW fuzzy logic profile used for the piston controller.
- [3] PistonFuzzyControlProfile1Desc.pdf:
Description of Fuzzy Profile 1, including its inputs, outputs, membership functions, and rules.
- [4] FuzzyProfile1_SimRun.xlsx: Simulated baseline run for Fuzzy Profile 1.
- [5] Physical test Run 1_5-1.xlsx: Recorded real piston test run.
- [6] Physical test Run 2_5-1.xlsx: Recorded real piston test run.
- [7] Physical test Run 3_5-1.xlsx: Recorded real piston test run.
- [8] Physical test Run 4_5-1.xlsx: Recorded real piston test run.
- [9] Physical test Run 5_5-1.xlsx: Recorded real piston test run.
- [10] Summary of Test Runs Profile1.xlsx: Summary of extracted values from the real and simulated runs.
- [11] Prompt1.txt: Initial engineering-analysis instructions reproduced verbatim in Appendix B.
- [12] Prompt2.txt: Controller-optimization request reproduced verbatim in Appendix C.

APPENDIX B

PROMPT 1 (VERBATIM)

You are an engineering analysis assistant specialized in control systems, system identification, and fuzzy logic control applied to hydraulic actuators.

This project focuses on an empirically modeled hydraulic piston system governed by:

$a = K_{upper} \cdot u_{upper} - K_{lower} \cdot u_{lower} - c \cdot v$ where valve commands produce acceleration, which is numerically integrated to velocity and position. The model replaces pressure-based force dynamics with experimentally calibrated gains.

Your role is to provide rigorous, technically grounded analysis of this system and its fuzzy controller.

When responding, ALWAYS:

1. Prioritize governing dynamics
 - Interpret the system as a second order actuator
 - Explain cause-effect relationships (input → acceleration → velocity → position)
 - Identify stability characteristics, damping behavior, and transient response
2. Analyze the model critically
 - Distinguish between physical reality and empirical approximation
 - Identify missing dynamics (pressure, flow, nonlinearities, delays, friction)
 - Evaluate the validity and limitations of calibrated gains
3. Evaluate control behavior
 - Analyze how fuzzy inputs (error, error rate) map to valve outputs
 - Identify aggressiveness, smoothness, and asymmetry
 - Detect oscillation sources, overshoot causes, and inefficiencies
4. Use engineering metrics where applicable
 - Tracking error
 - Rise time
 - Overshoot
 - Settling time
 - Steady-state error
 - Energy/valve usage
5. Be diagnostic, not just descriptive
 - Identify WHY behavior occurs, not just WHAT happens
 - Trace issues back to gains, damping, controller logic, or model structure
6. Provide actionable recommendations
 - Suggest improvements to fuzzy rules, membership functions, or scaling
 - Recommend model refinements (e.g., nonlinear gains, dead zones, asymmetric damping)

- Highlight trade-offs (stability vs responsiveness, accuracy vs simplicity)

7. Maintain technical rigor

- Use precise engineering language
- Avoid vague statements
- Make assumptions explicit when data is missing

8. When data is provided:

- Infer system behavior from trends (not just raw values)
- Compare across runs
- Identify inconsistencies or anomalies

9. When data is NOT provided:

- Clearly state assumptions
- Base conclusions on the given model structure

Do NOT:

- Give generic explanations of fuzzy logic or control theory unless directly relevant
- Treat the model as physically complete
- Ignore asymmetry between K_{upper} and K_{lower} - Overlook damping and integration effects

Primary objective:

Use the available data and model to improve fuzzy control performance and system behavior.

Focus on real engineering insight, not surface-level summaries.

APPENDIX C

PROMPT 2 (VERBATIM)

This is the correct strategy - Main technical implication: this controller can react to position error, but it has no direct awareness of piston velocity. So overshoot, oscillation, and settling behavior must be managed indirectly through the spacing of the distance membership functions, output duty-cycle levels, damping in the empirical plant model, and the asymmetry between K_{upper} and K_{lower} .

Going forward, evaluate every fuzzy profile against: $a = K_{upper} * u_{upper} - K_{lower} * u_{lower} - c * v$ and connect controller

behavior back to physical-style effects: acceleration authority, damping, integration drift, valve usage, rise time, overshoot, cumulative error, and steady-state tracking.

Your goals are as follows

Draft an improved fuzzy logic profile which aims to beat the current one in terms of

- retract max velocity
- extend max velocity
- rise time
- overshoot
- average absolute error
- cumulative absolute error
- settling behavior

Note: The fuzzy system cannot produce the high command you think it can, because the output membership centroid is too low. It is using Center of Area and try to ensure that the valve can reach fully open.

Now explain the configuration of an optimized fuzzy profile logic to eclipse the previous one in terms of performance. Produce the membership functions for each of the 3 variables including: Name, Shape, and the 4 points for each individual membership within the variables respectfully.